

5-Fold Cross Validation Evaluation Report (Ensemble + Calibration)

Started: 2026-04-14 14:50:58

Finished: 2026-04-14 15:42:28

Model Architecture: DenseNet121

Phase 1 LR: 0.01 | Phase 2 LR: 1e-06

Aggregate Performance across 5 Folds (Individual Optimized Thresholds):

Average Accuracy: 0.8814 ± 0.0048

Average Sensitivity: 0.9059 ± 0.0177

Average Specificity: 0.8561 ± 0.0273

Average AUC: 0.9502 ± 0.0011

CALIBRATED ENSEMBLE TEST METRICS:

Forced Decision Threshold: 0.5000

Accuracy : 0.8488

Precision : 0.9461

Sensitivity (Recall) : 0.7451

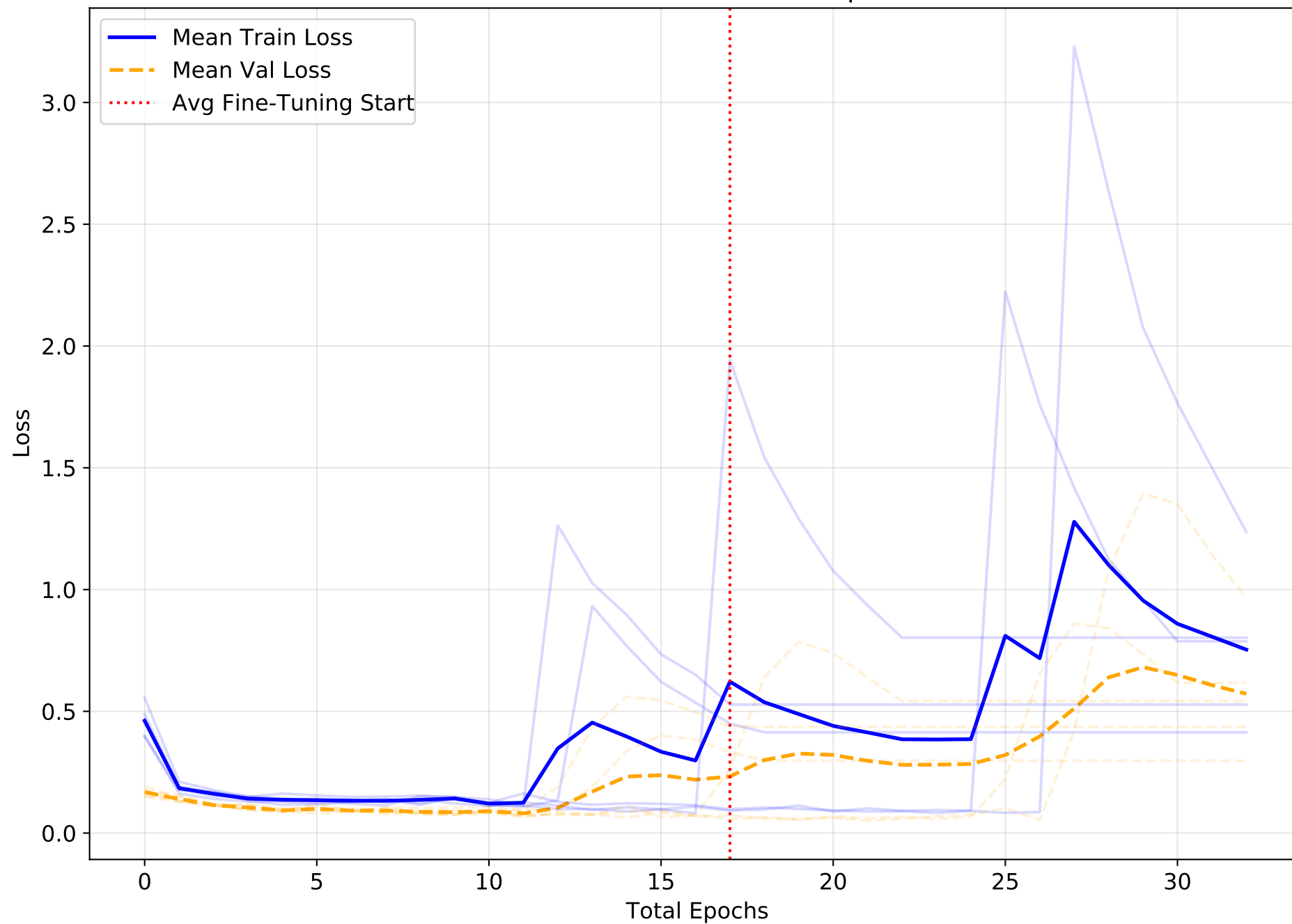
Specificity : 0.9561

Ensemble AUC : 0.9517

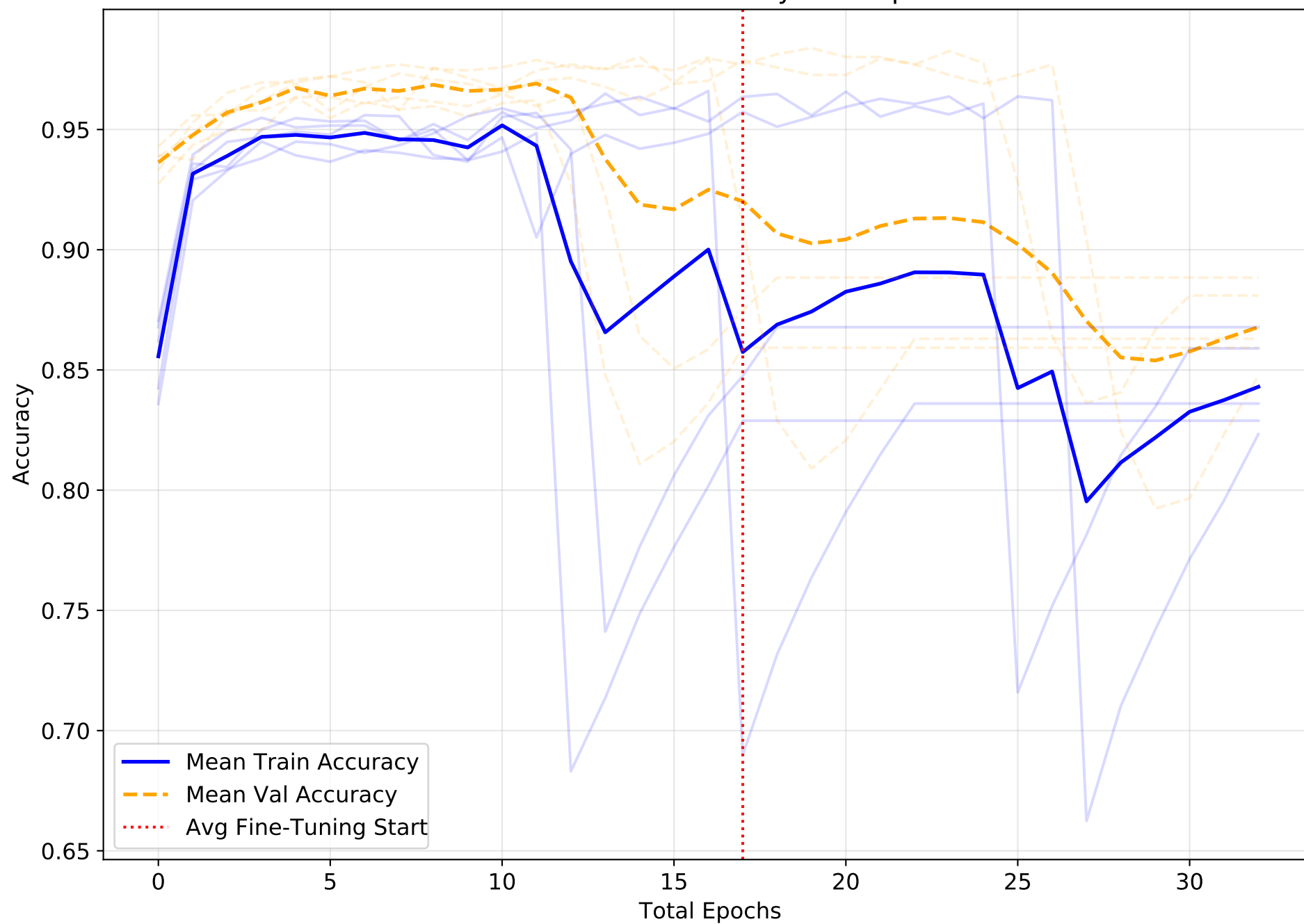
Classification Report (at threshold 0.5000):

	precision	recall	f1-score	support
Normal	0.78	0.96	0.86	296
Cataract	0.95	0.75	0.83	306
accuracy			0.85	602
macro avg	0.86	0.85	0.85	602
weighted avg	0.87	0.85	0.85	602

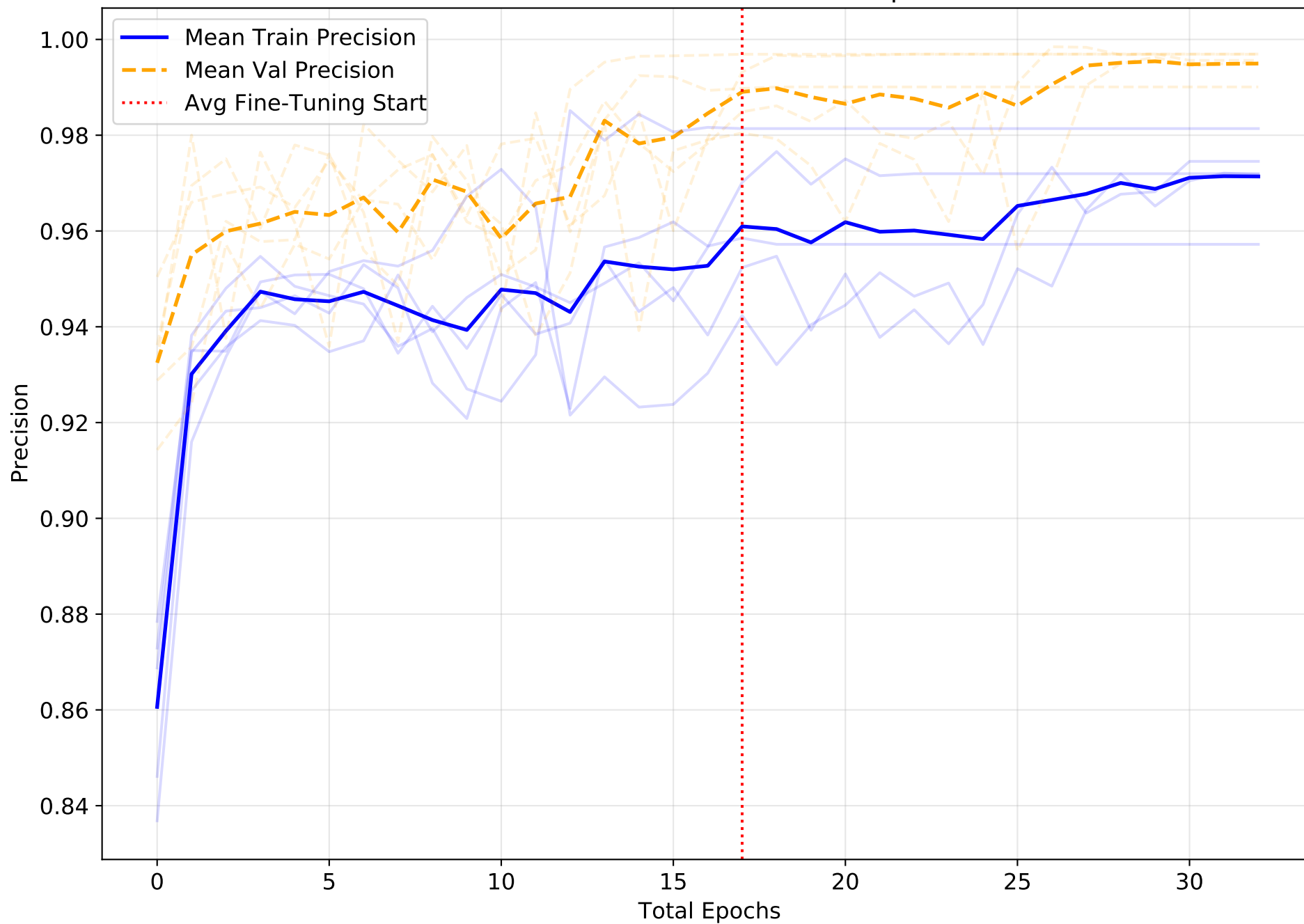
Cross-Validated Loss Over Epochs



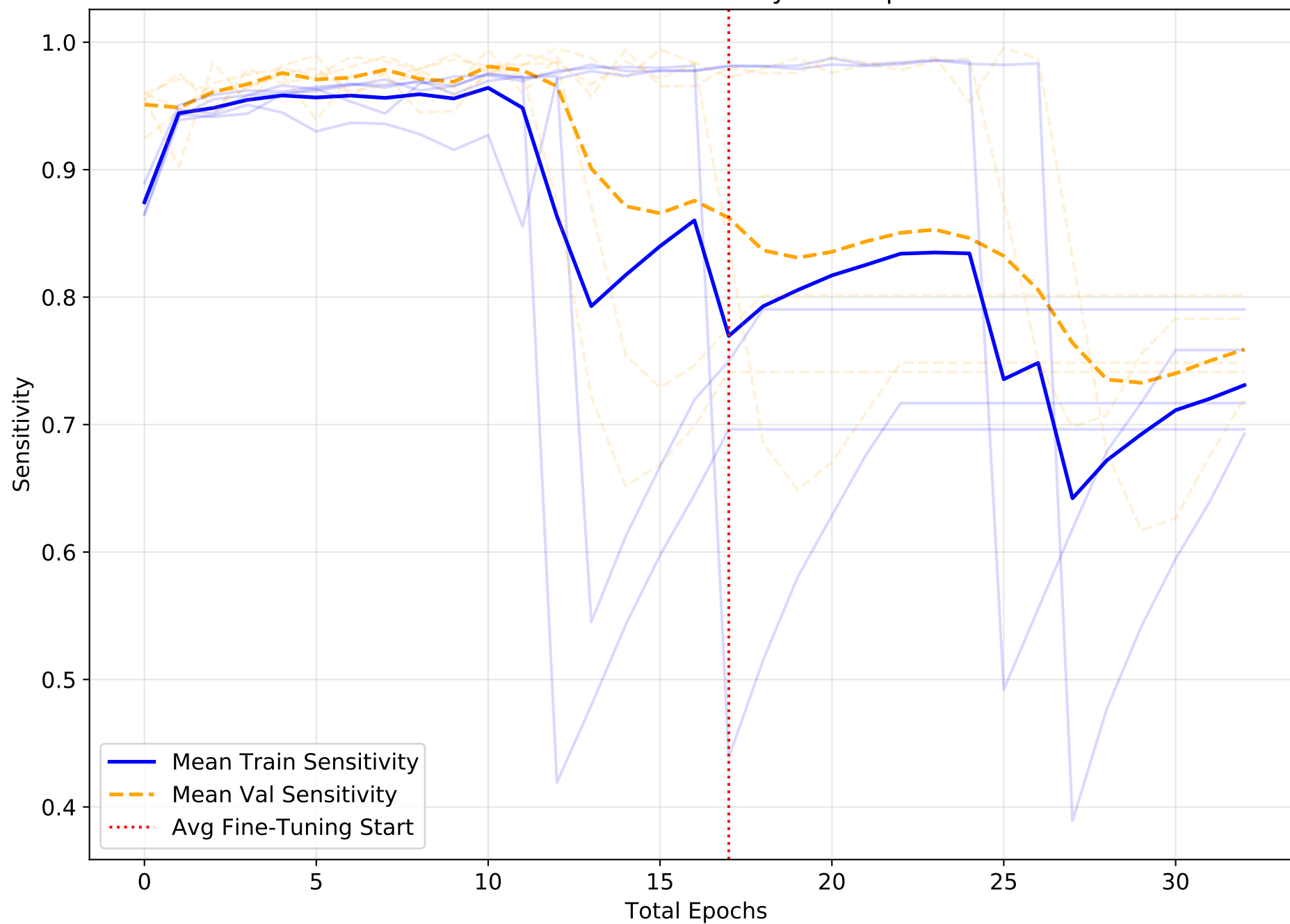
Cross-Validated Accuracy Over Epochs



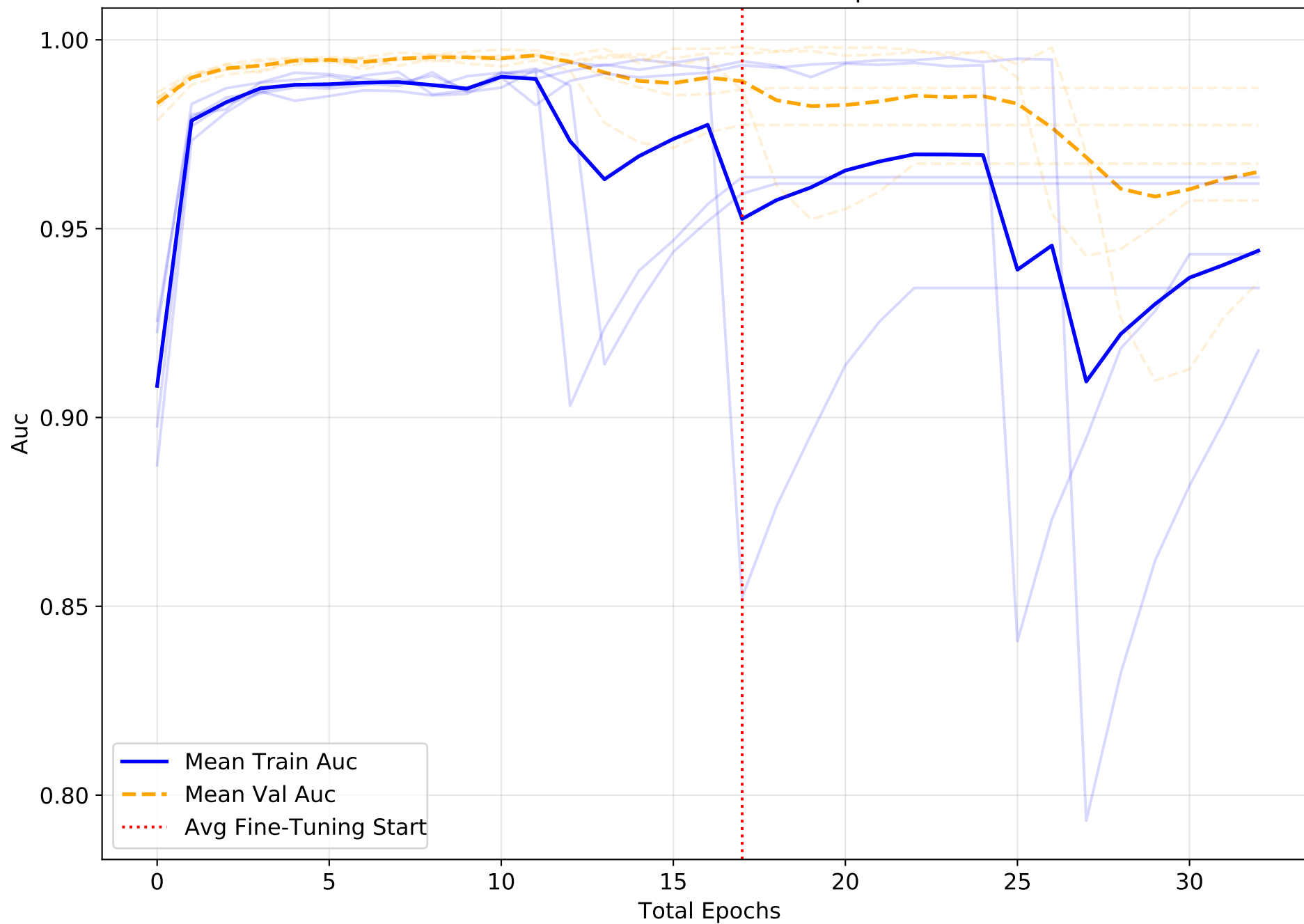
Cross-Validated Precision Over Epochs



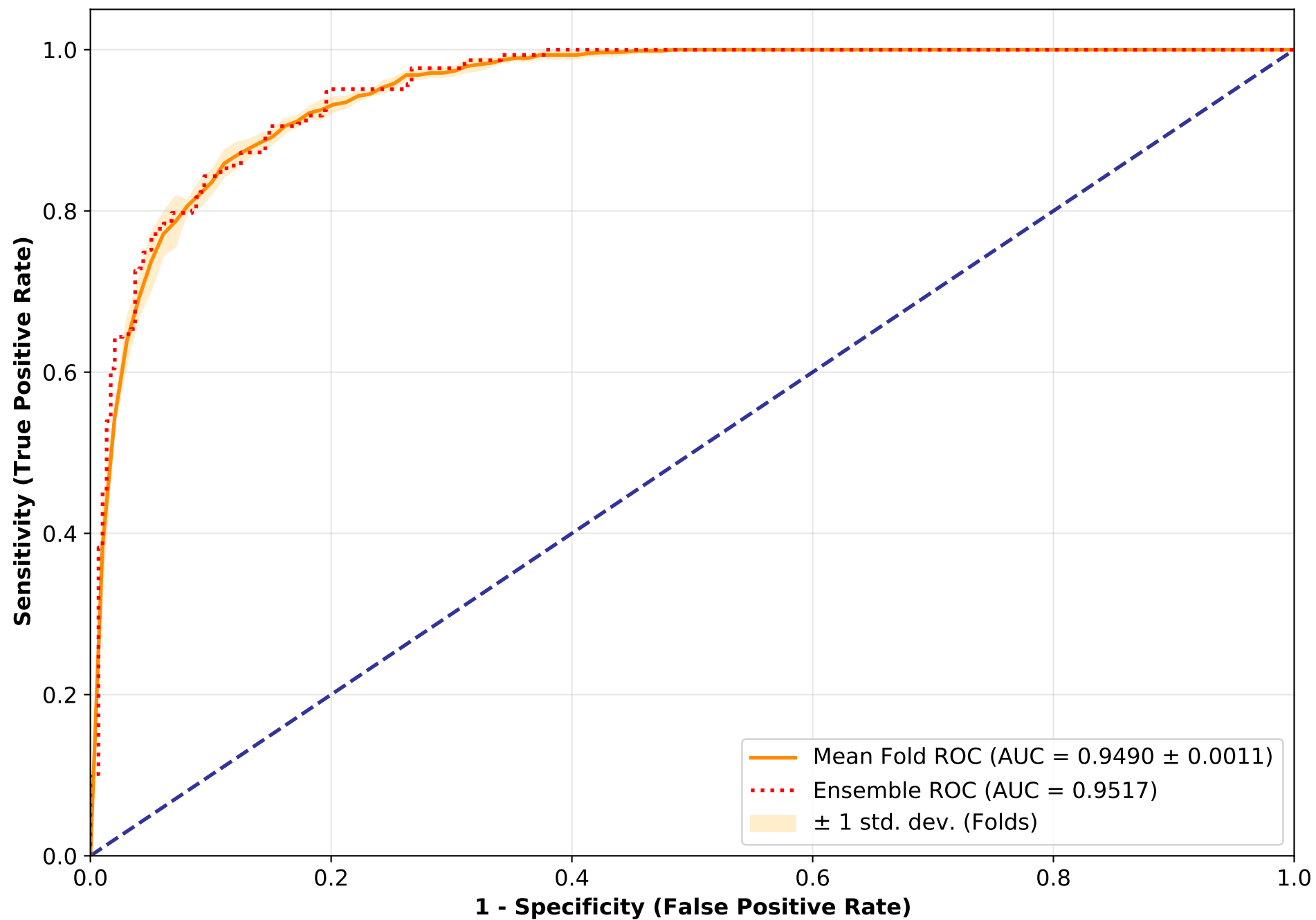
Cross-Validated Sensitivity Over Epochs



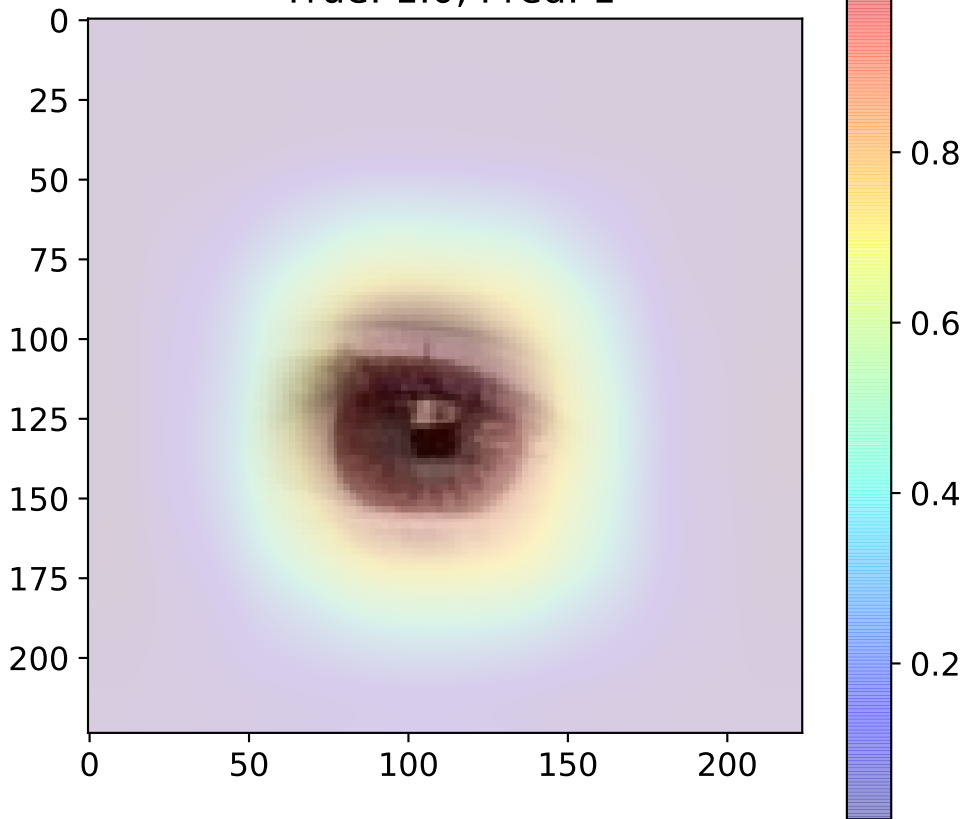
Cross-Validated Auc Over Epochs



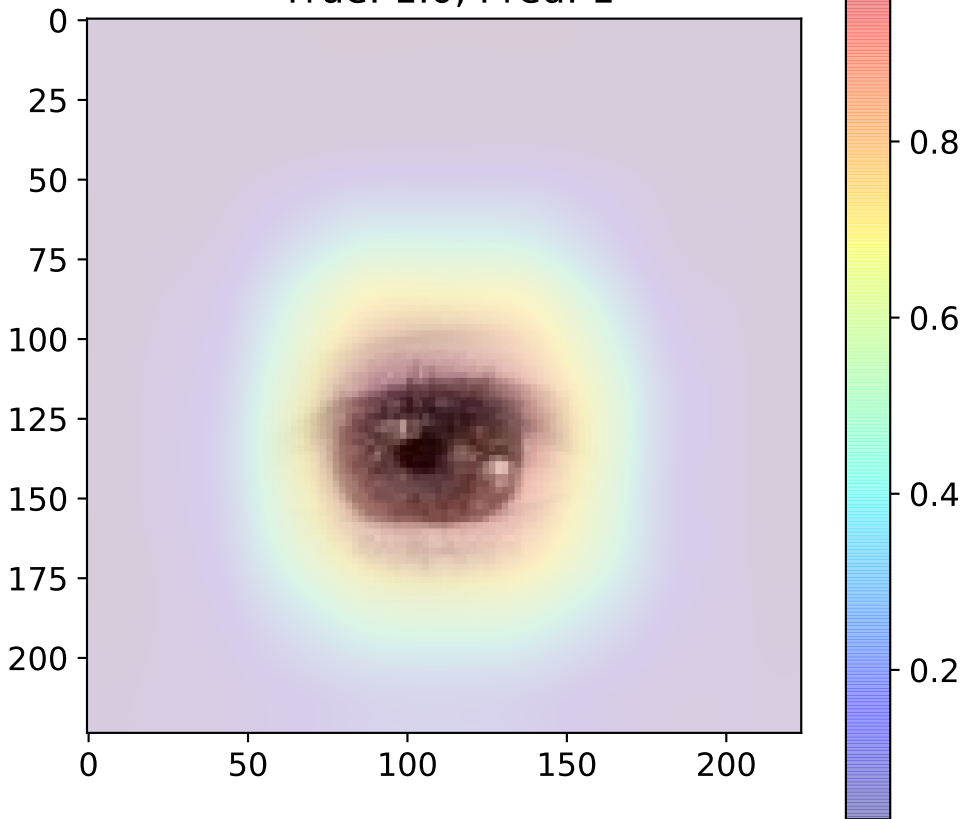
Cross-Validated ROC Curve - DenseNet121



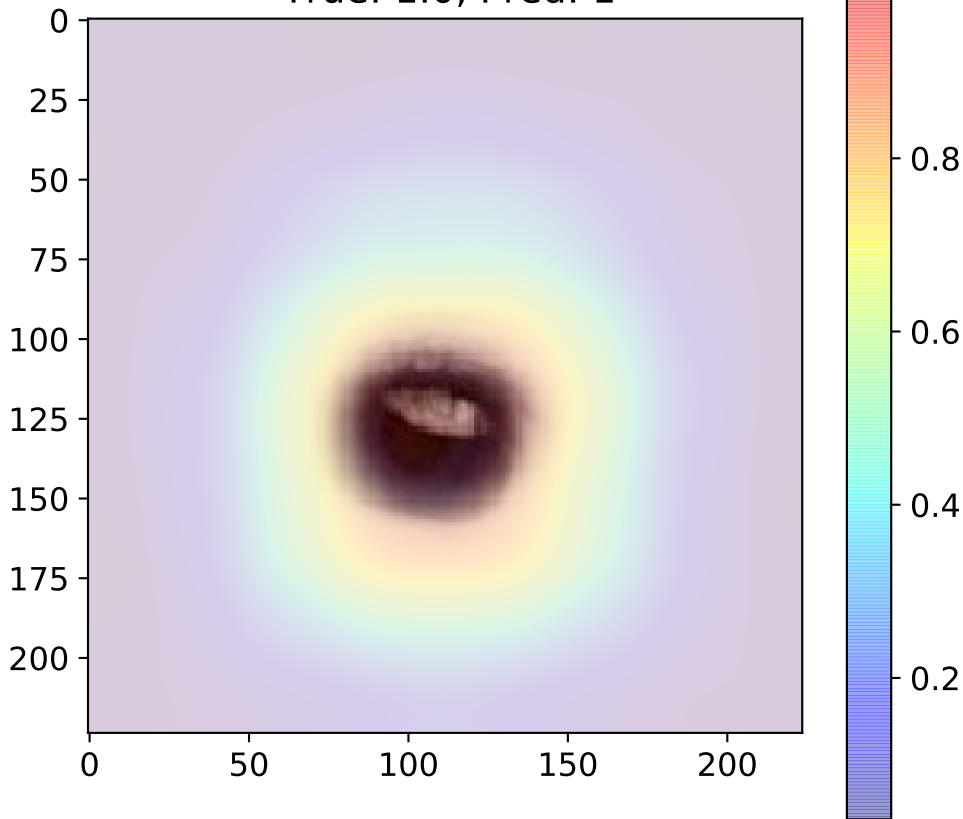
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



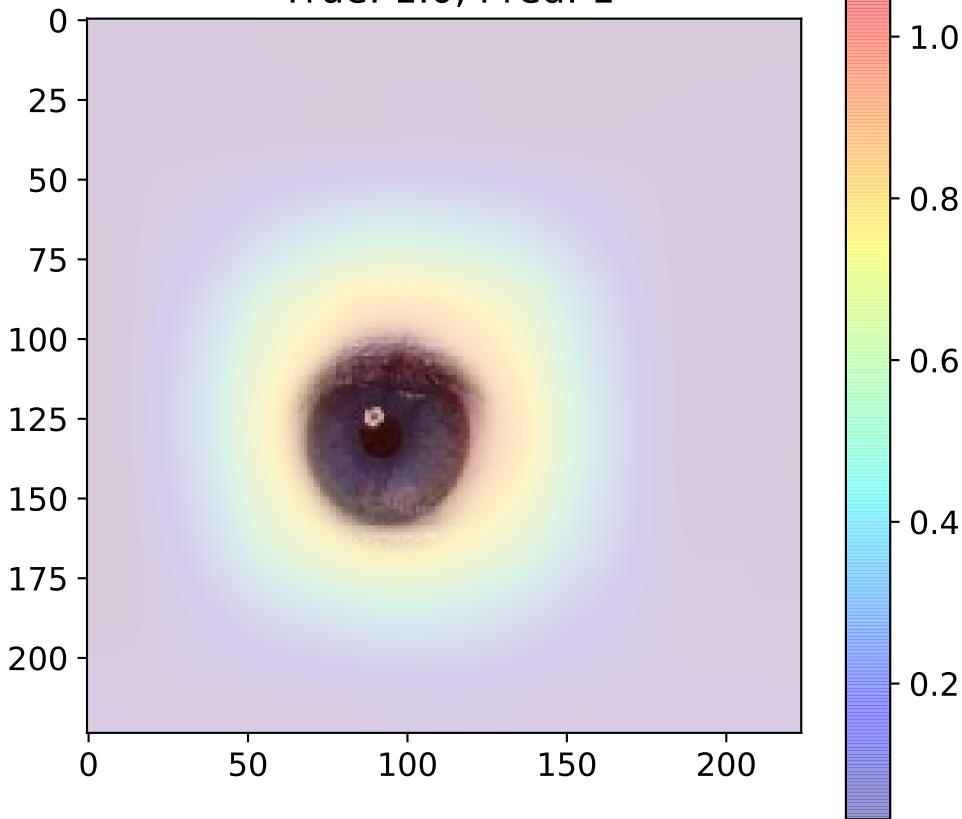
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



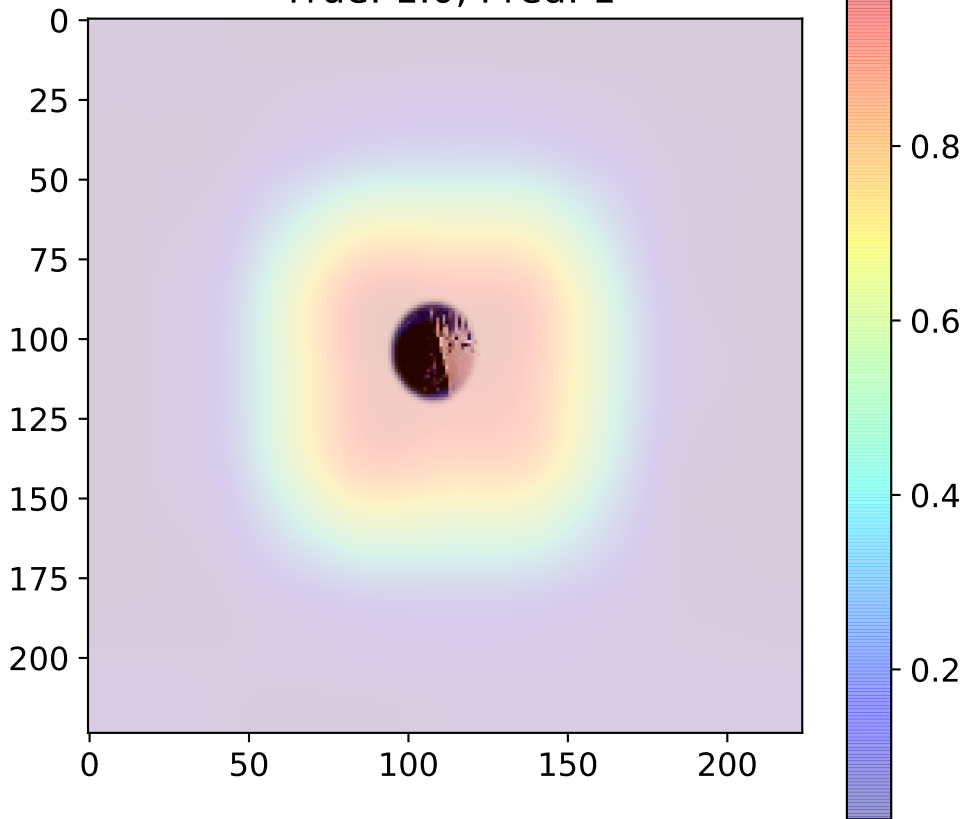
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



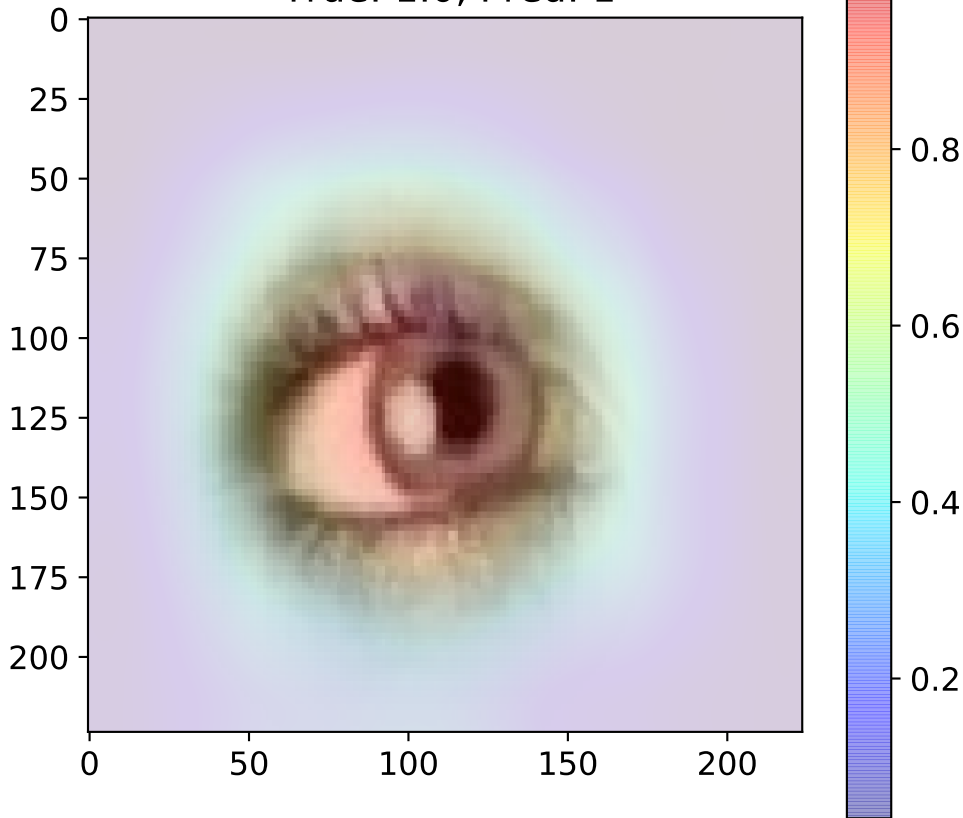
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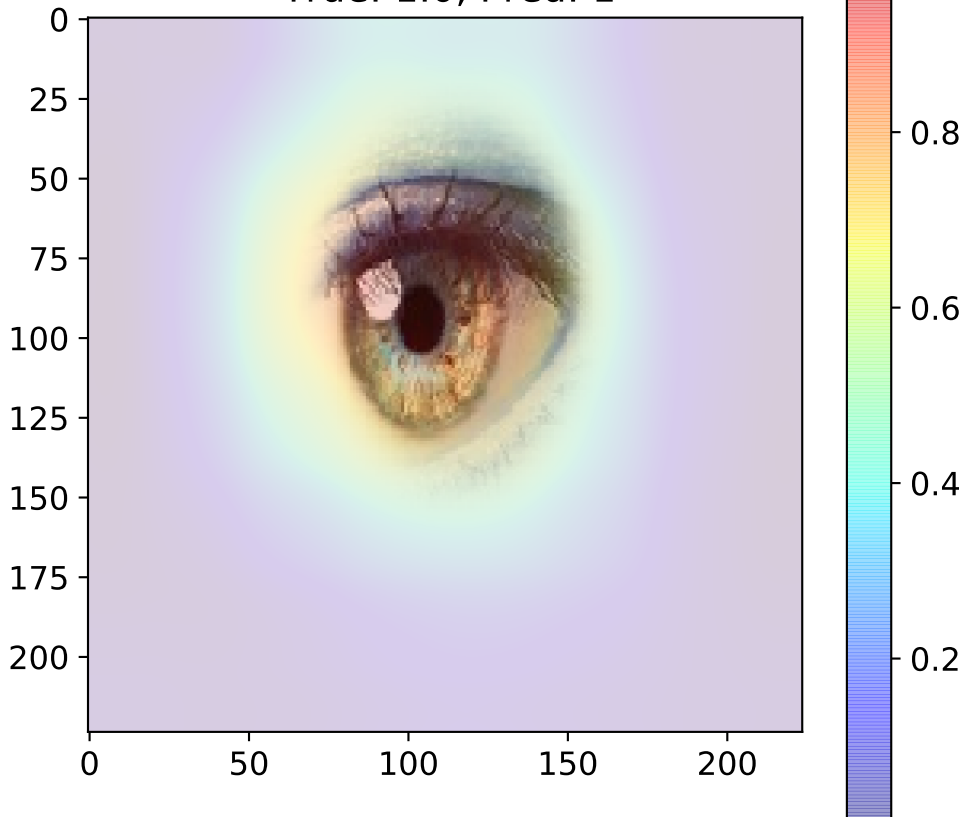
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True: 1.0, Pred: 1



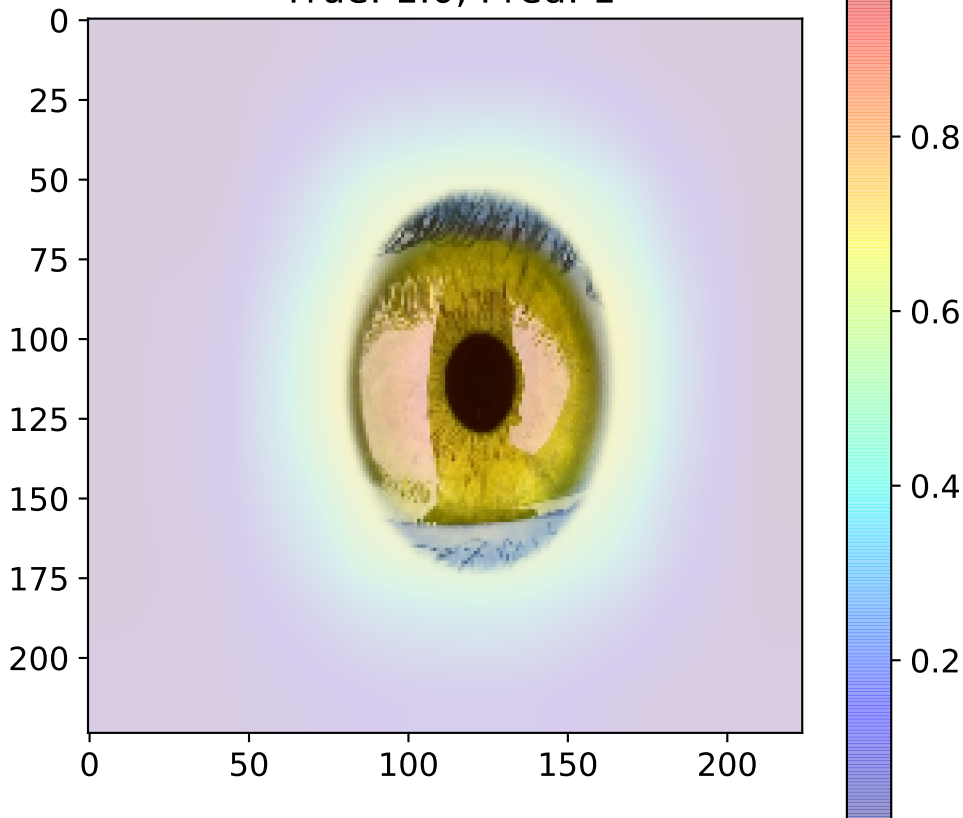
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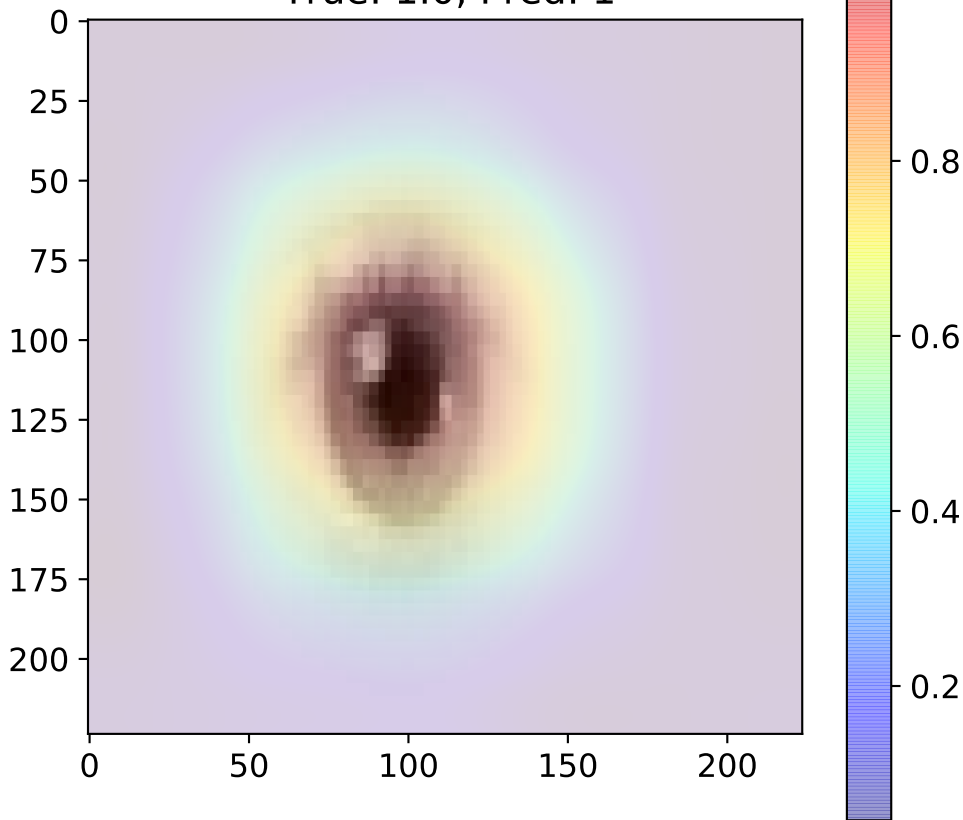


Correctly Classified (Best Fold)
True: 1.0, Pred: 1

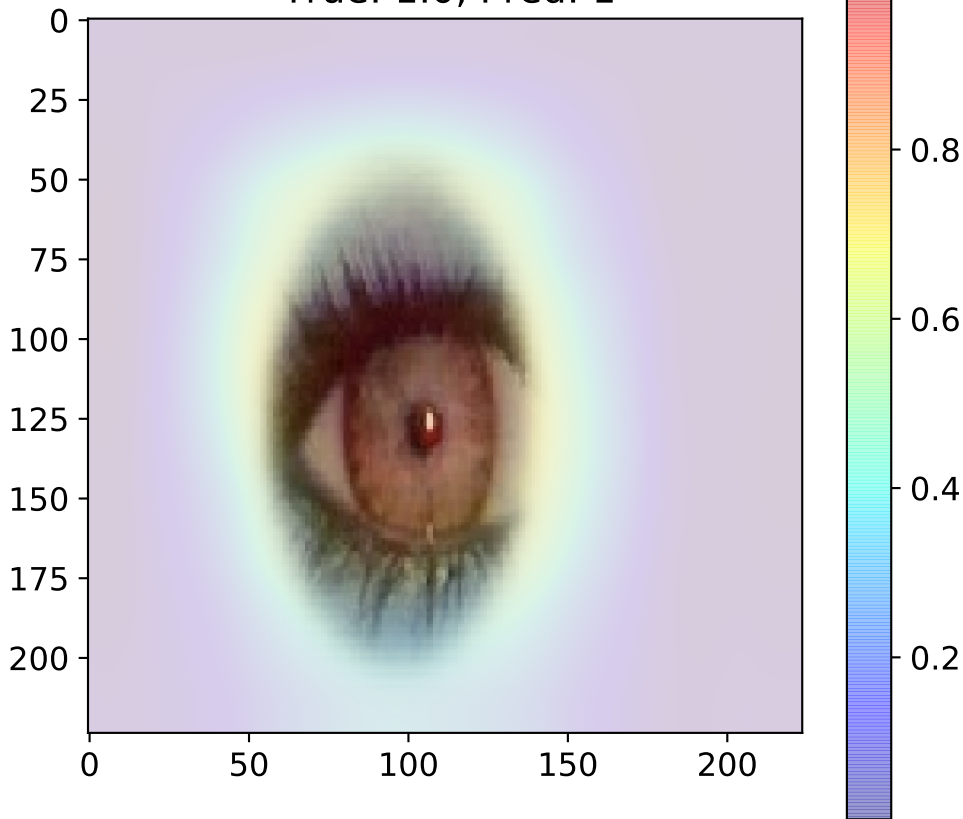


Correctly Classified (Best Fold)

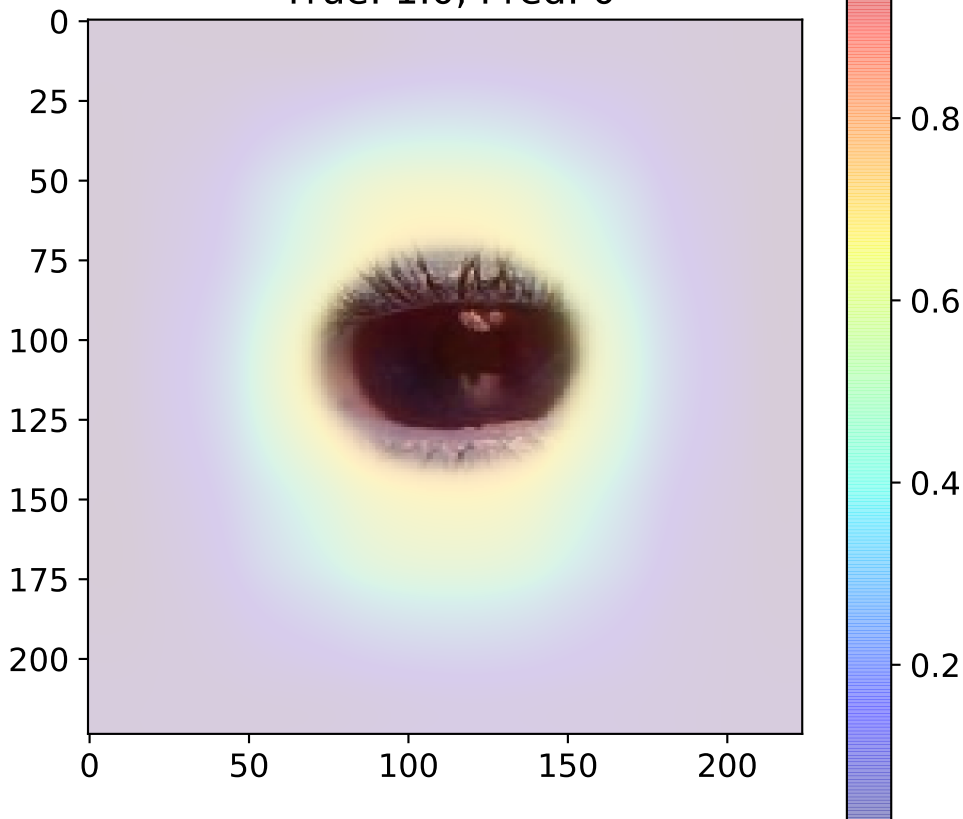
True: 1.0, Pred: 1



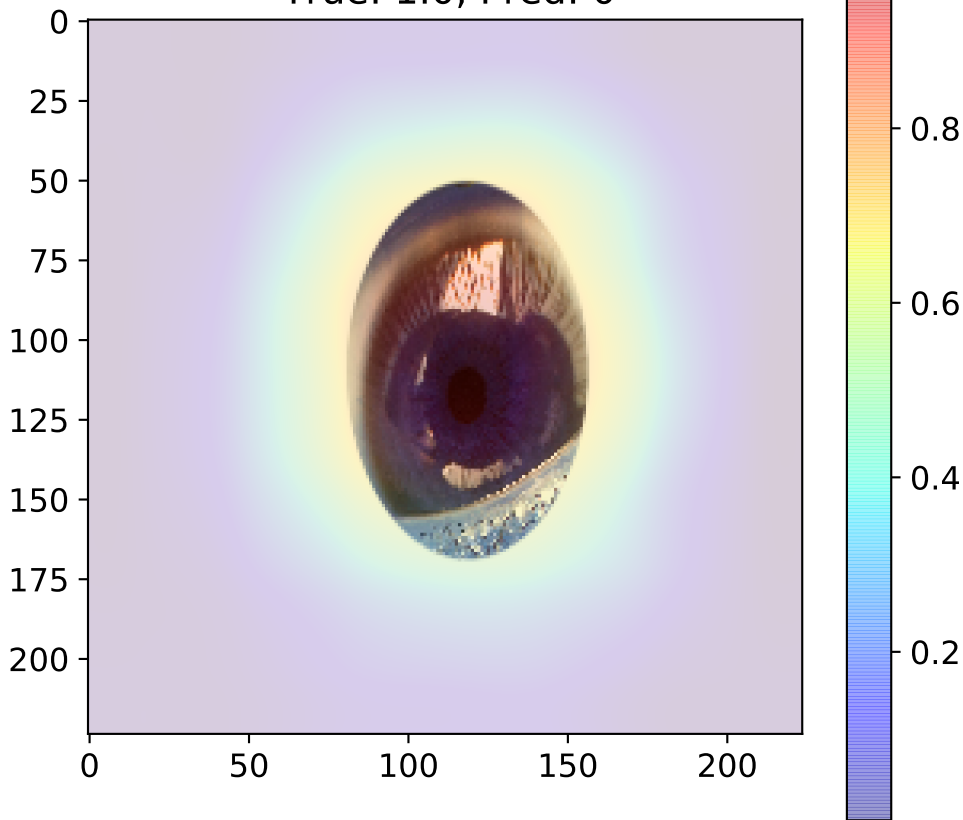
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



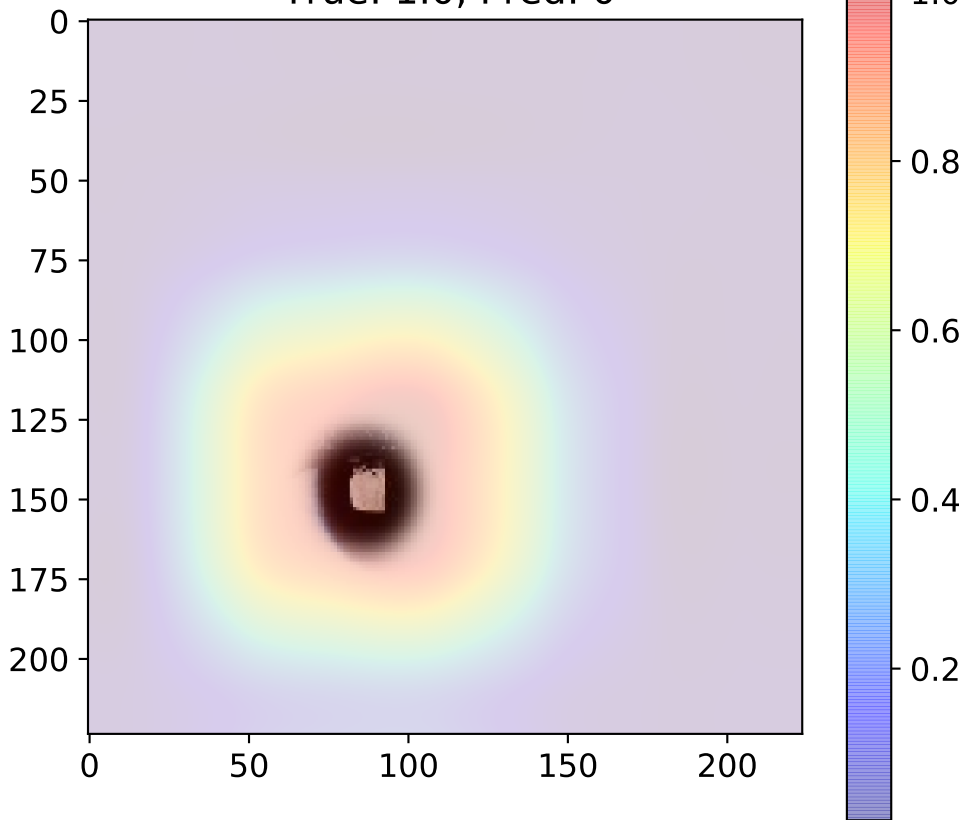
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



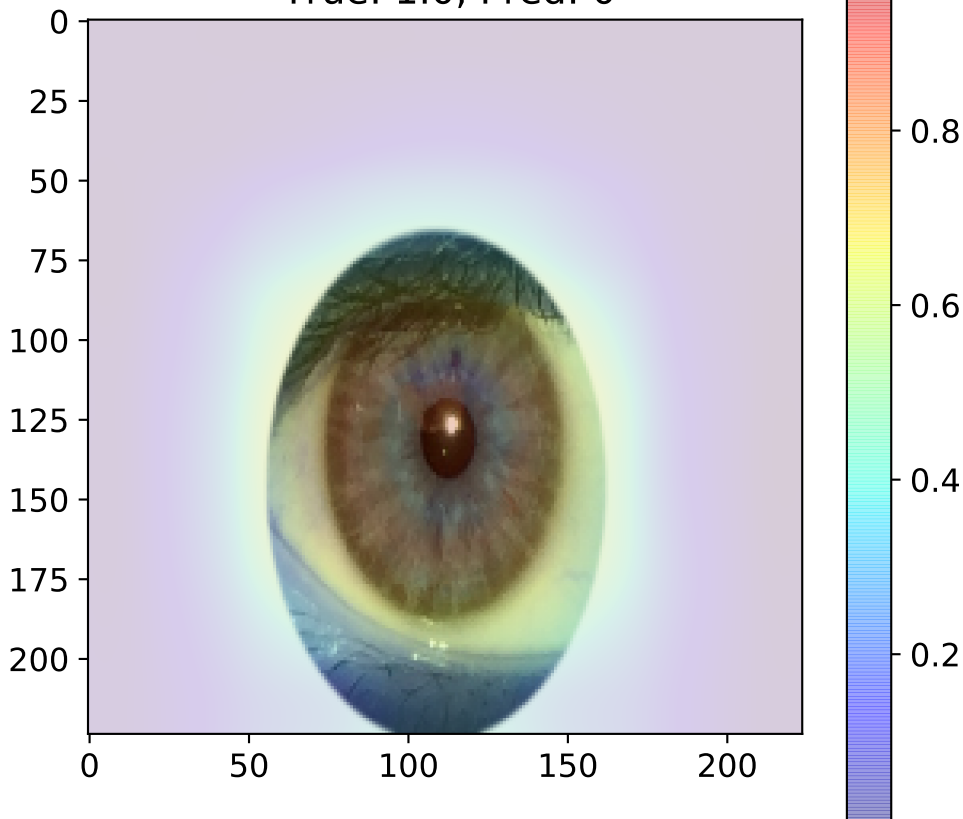
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



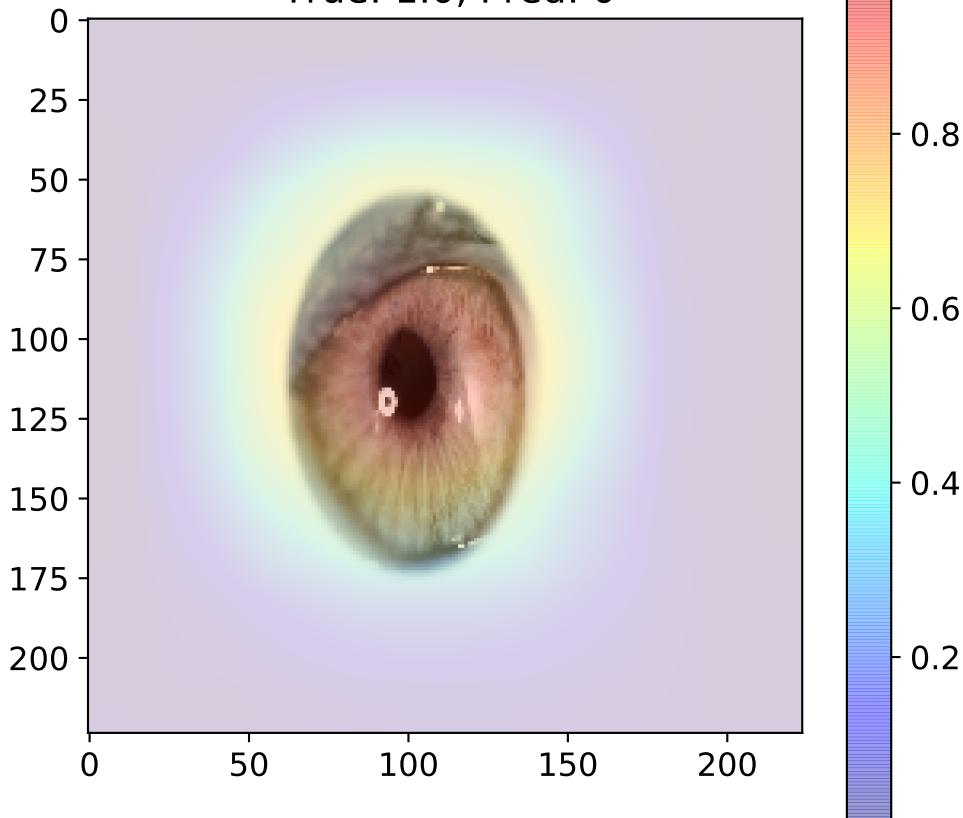
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



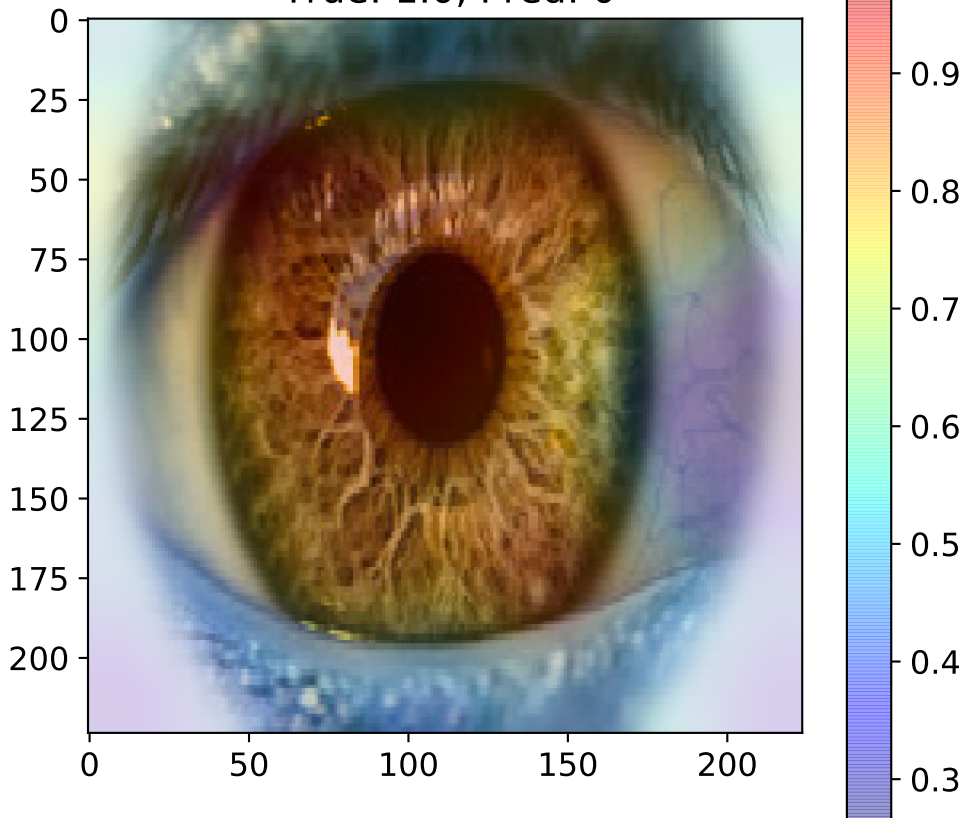
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



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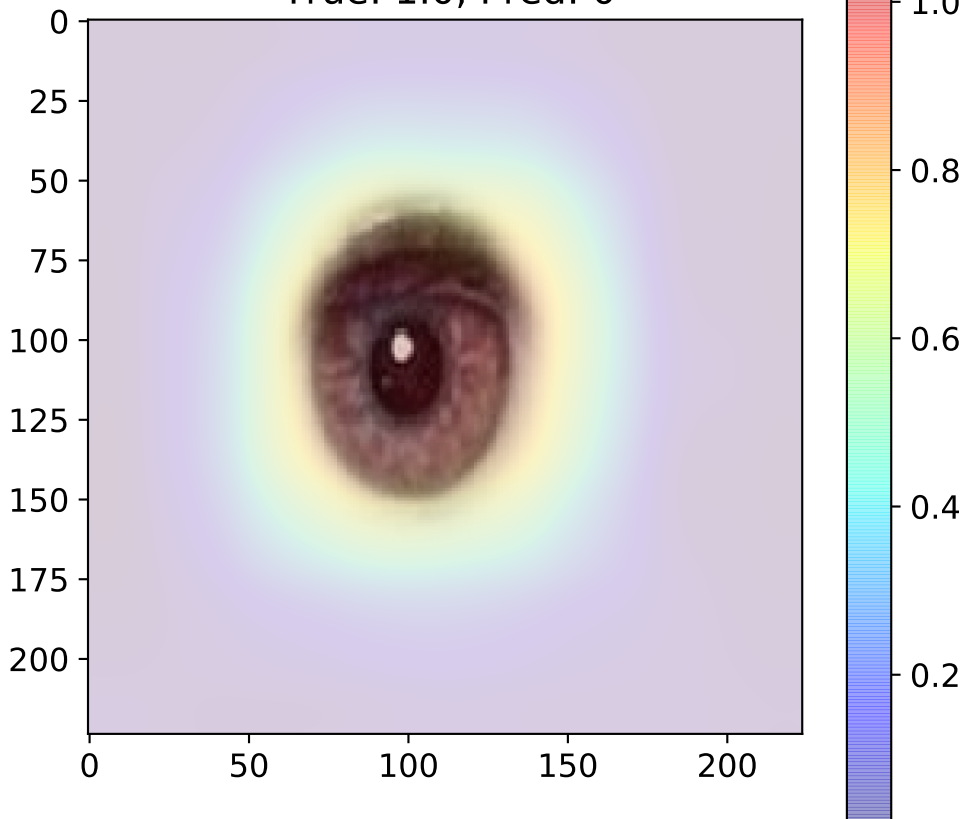


Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



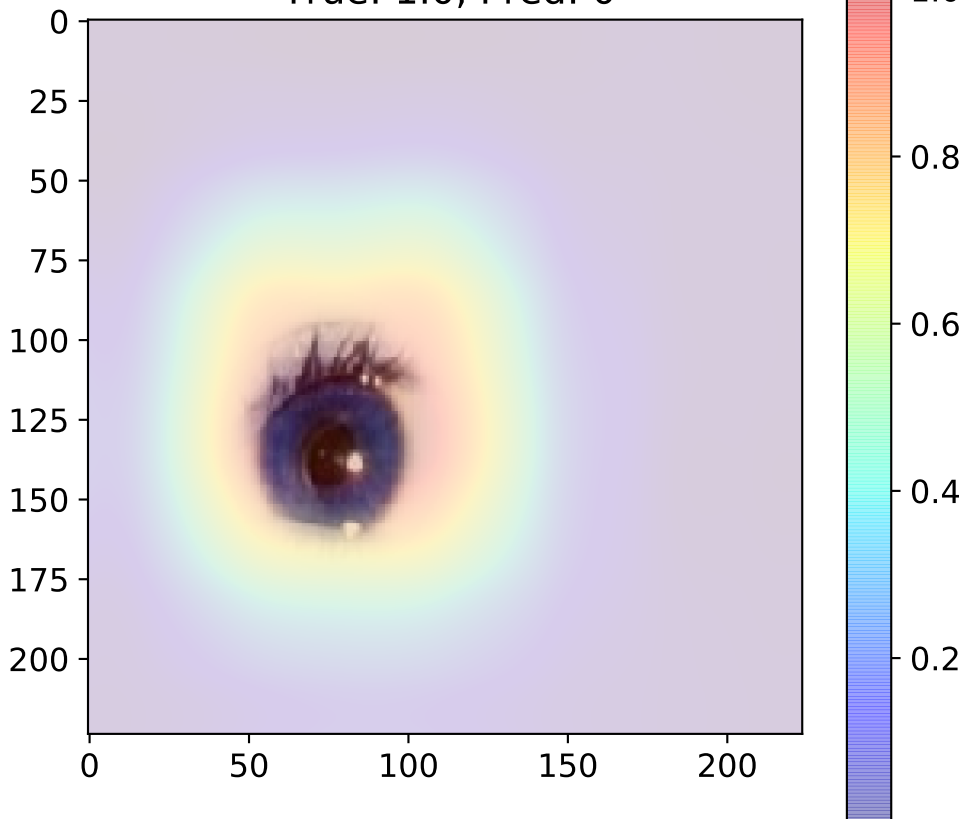
Incorrectly Classified (Best Fold)

True: 1.0, Pred: 0

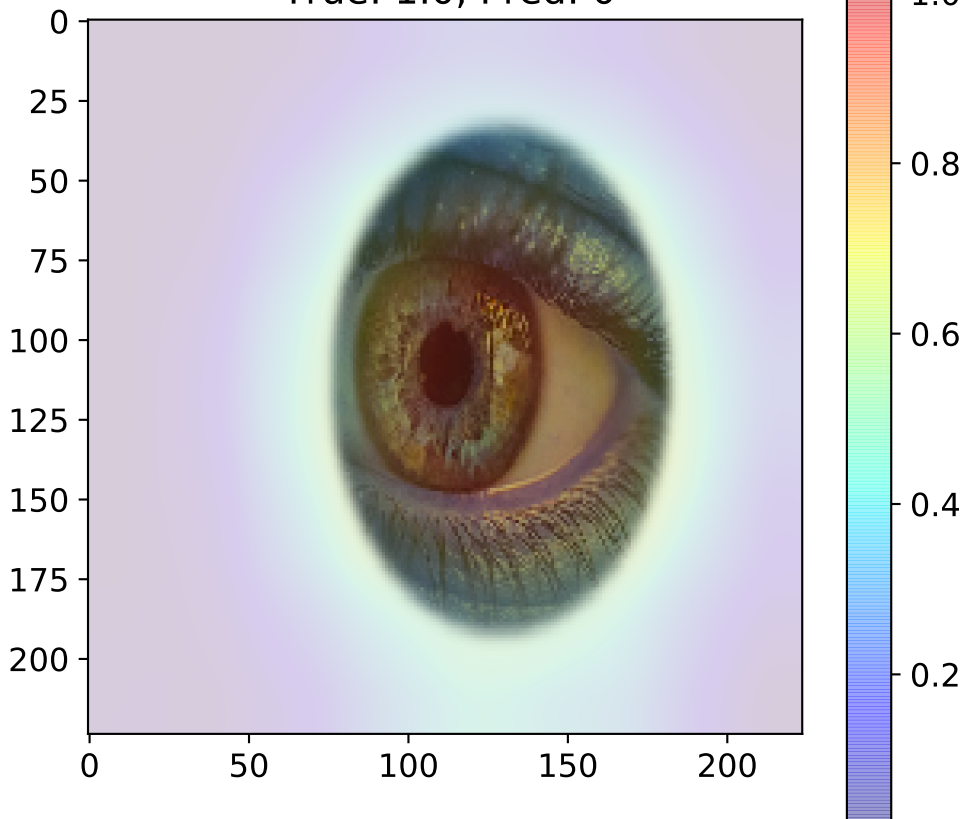


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